★ Estimation

Since Y is binary and rare

$$\log(E(Y|A,M,C)) = \theta_0 + \theta_1 A + \theta_2 M + \theta_c C$$

$$Q(a,a^*) = \sum_c \sum_m E(Y|M=m, A=a, C=c) P(M=m|A=a^*, C=c) P(C=c)$$

$$= \sum_c \sum_m \{ e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c c} \} P(M=m|A=a^*, C=c) P(C=c)$$

$$= e^{\theta_0+\theta_1 a} \sum_c e^{\theta_c c} E(e^{\theta_2 M}|A=a^*, C=c) P(C=c)$$

if M is conti, $M \sim N(\mu_M, \delta^2)$

$$\mu_M = E(M|A=a^*, C=c) = \alpha_0 + \alpha_1 a^* + \alpha_c c$$

then $E(e^{\theta_2 M}|A=a^*, C=c) = e^{\mu_M\theta_2 + \frac{1}{2}\delta^2\theta_2^2}$

$$= e^{\theta_2(\alpha_0+\alpha_1 a^*+\alpha_c c) + \frac{1}{2}\delta^2\theta_2^2}$$

$$\Rightarrow Q(a,a^*) = e^{\theta_0+\theta_1 a} E\left[e^{\theta_2(\alpha_0+\alpha_1 a^*+\alpha_c c) + \frac{1}{2}\delta^2\theta_2^2}\right]$$

$$= e^{\theta_0+\theta_1 a + \theta_2(\alpha_0+\alpha_1 a^*) + \frac{1}{2}\delta^2\theta_2^2} E\left(e^{(\alpha_c\theta_2+\theta_c)c}\right)$$

$$NDE_{RR} = \frac{e^{\theta_0+\theta_1+\alpha_0\theta_2+\frac{1}{2}\delta^2\theta_2^2}}{e^{\theta_0+\alpha_0\theta_2+\frac{1}{2}\delta^2\theta_2^2}} = e^{\theta_1}$$

$$NIE_{RR} = e^{\alpha_1\theta_2}$$

if M is Binary

$$\text{logit}[P(M=1|A=a^*, C=c)] = \alpha_0 + \alpha_1 a^* + \alpha_c c$$

define $P(M=1|A=a^*, C=c) = \text{logit}^{-1}(\alpha_0+\alpha_1 a^*+\alpha_c c)$

$$= \pi(a^*, c, \alpha)$$

$$P(M=0|A=a^*, C=c) = 1 - \pi(a^*, c, \alpha)$$

then $E(e^{\theta_2 M}|A=a^*, C=c) = e^{\theta_2}\pi(a^*, c, \alpha) + e^0[1-\pi(a^*, c, \alpha)]$

$$= 1 + (e^{\theta_2}-1)\pi(a^*, c, \alpha)$$

- If outcome is rare and M follows a linear regression, the differece method is approximately equal to the $\log(NIE_{RR})$

Let T be time-to-event outcome.
(Measure the amount time from start pt until a
a specific event occurs)
A, M, C remain unchanged.

★ Recap.

$T \sim f(t)$

$S(t) = P(T>t) = 1-F(t)$ is the survival fn at time t.

$\Updownarrow$

$S(t|c) = P(T>t|c)$

$$-\frac{dS(t)}{dt} = f(t)$$

$$\lambda(t) = \lim_{\Delta t \to 0^+} \frac{P(t \le T < t+\Delta t \mid T \ge t)}{\Delta t} = \frac{f(t)}{S(t)} = -\frac{d \ln S(t)}{dt}$$

$$\Rightarrow \quad e^{-\Lambda(t)} = e^{-\int_0^t \lambda(s)ds} = S(t)$$

hazard at time t
Instantaneous rate of the event conditional on $T \ge t$

★ Counterfactuals

$T(a)$ is counterfactual event time if $A=a$
$M(a)$ mediator $A=a$
$T(a,m)$ event time $A=a, M=m$

$T(a, M(a^*))$ is nested counterfactual event time if $A=a$ and mediator had been set the level it would had been $A=a^*$

★ There are multiple ways or scales by which we might decompose a total effect comparing exposure levels a and a^* into direct and indirect effect.

Ex. log hazard.

No. : 50-2

Subject : .... Date : ..../......../........

$$\log\{\lambda_{T(a)}(t)\} - \log\{\lambda_{T(a^*)}(t)\} = \left[\log\{\lambda_{T(a,M(a))}(t)\} - \log\{\lambda_{T(a,M(a^*))}(t)\}\right] + \left[\log\{\lambda_{T(a,M(a^*))}(t)\} - \log\{\lambda_{T(a^*,M(a^*))}(t)\}\right]$$

$$\Updownarrow$$

$$\frac{\lambda_{T(a)}(t)}{\lambda_{T(a^*)}(t)} = \frac{\lambda_{T(a,M(a))}(t)}{\lambda_{T(a,M(a^*))}(t)} \times \frac{\lambda_{T(a,M(a^*))}(t)}{\lambda_{T(a^*,M(a^*))}(t)}$$

We have that

$$\lambda_{T(a,M(a^*))}(t) = \frac{f_{T(a,M(a^*))}(t)}{S_{T(a,M(a^*))}(t)}$$

$$\begin{aligned} f_{T(a,M(a^*))}(t) &= \int f_{T(a,M(a^*))}(t \mid M(a^*)=m)\, dF_{M(a^*)}(m) \\ &= \int f_{T(a,m)}(t \mid M(a^*)=m)\, dF_{M(a^*)}(m) \\ &= \int f_{T(a,m)}(t \mid A=a, M=m)\, dF_M(m \mid a^*) \\ &= \int f_T(t \mid a,m)\, dF_M(m \mid a^*) \end{aligned}$$

$$S_{T(a,M(a^*))}(t) = \int S_T(t \mid a,m)\, dF_M(m \mid a^*) = \int e^{-\Lambda(t \mid a,m)}\, dF_M(m \mid a^*)$$

Consider Proportional Hazard Model

$$\lambda_T(t \mid a,m) = \underbrace{\lambda_T(t \mid 0,0)}_{\text{underlying hazard}} e^{\theta_1 a + \theta_2 m}$$

and mediator model

$$E(M \mid A) = \beta_0 + \beta_1 A$$

then

$$f_{T(a,M(a^*))}(t) = \int \lambda_T(t \mid 0,0) e^{\theta_1 a + \theta_2 m} e^{-[\Lambda(t \mid 0,0) e^{\theta_1 a + \theta_2 m}]}\, dF_M(m \mid a^*)$$

$$S_{T(a,M(a^*))}(t) = \int e^{-[\Lambda(t \mid 0,0) e^{\theta_1 a + \theta_2 m}]}\, dF_M(m \mid a^*)$$

$$\begin{aligned} \Rightarrow \lambda_{T(a,M(a^*))} &\approx \lambda_T(t \mid 0,0)\, e^{\theta_1 a + \theta_2(\beta_1 + \beta_2 a^*) + \frac{1}{2}\theta_2^2 \sigma^2} \\ &= \left(\lambda_T(t \mid 0,0)\, e^{\theta_2 \beta_1 + \frac{1}{2}\theta_2^2 \sigma^2}\right) e^{\theta_1 a + \theta_2 \beta_2 a^*} \end{aligned}$$

$$\frac{\lambda_{T(1,M(1))}(t)}{\lambda_{T(1,M(0))}(t)} = e^{\theta_2 \beta_2}, \quad \frac{\lambda_{T(1,M(0))}(t)}{\lambda_{T(0,M(0))}(t)} = e^{\theta_1}$$

★ Differece method is lower bdd for NIE_{OR} (Underestimate)

$$\begin{pmatrix} \text{logit}(p) = x \\ \Updownarrow \\ p = \frac{e^x}{1+e^x} \end{pmatrix} \qquad \text{logit}^{-1}(\theta_0+\theta_1 a+\theta_2 m+\theta_c C) = \frac{e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}}{1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}}$$

$$= P(Y=1 \mid M=m, A=a, C=c)$$

$$\mathbb{E}[Y(a, M(a^*)) \mid C] = \int_m \mathbb{E}(Y \mid M=m, A=a, C=c)\, f_{M|A,C}(m \mid a^*, c)\, dm$$

$$= P(Y(a, M(a^*)) = 1 \mid C) = \phi(a, a^*)$$

$$NDE_{OR} = \left[\frac{\phi(a,a^*)}{1-\phi(a,a^*)}\right] \Big/ \left[\frac{\phi(a^*,a^*)}{1-\phi(a^*,a^*)}\right]$$

Let $A = 1-\phi(a,a^*) = \int_m \frac{1}{1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}} f_{M|A,C}(m \mid a^*, c)\, dm$

$$= \mathbb{E}_{M|a^*,c}\left(\frac{1}{1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}}\right)$$

Similar

$$B = \mathbb{E}_{M|a^*,c}\left(\frac{1}{1+e^{\theta_0+\theta_1 a^*+\theta_2 m+\theta_c C}}\right)$$

$$NDE_{OR} = \left(\frac{1}{A}-1\right) \Big/ \left(\frac{1}{B}-1\right) \qquad \begin{pmatrix} NDE_{OR} \le e^{\theta_1(a-a^*)} \\ e^{\theta_1(a-a^*)}\left(\frac{1}{B}-1\right) - \left(\frac{1}{A}-1\right) \ge 0 \end{pmatrix}$$

Consider

$$e^{\theta_1(a-a^*)}\left(\frac{1}{B}-1\right) - \left(\frac{1}{A}-1\right)$$

$$= \frac{Ae^{\theta_1(a-a^*)} - B}{AB} - \left(e^{\theta_1(a-a^*)} - 1\right) = (\Delta)$$

$$Ae^{\theta_1(a-a^*)} - B = \mathbb{E}_{M|a^*,c}\left(\frac{e^{\theta_1(a-a^*)}[1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}] - [1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C}]}{(1+e^{\theta_0+\theta_1 a+\theta_2 m+\theta_c C})(1+e^{\theta_0+\theta_1 a^*+\theta_2 m+\theta_c C})}\right)$$

$$= \left[e^{\theta_1(a-a^*)} - 1\right] C$$

$$(\Delta) = \left(\frac{C}{AB} - 1\right)\left(e^{\theta_1(a-a^*)} - 1\right)$$

Lemma 1. Let f and g both non-decreasing

$Cov(f(x), g(x)) \ge 0$

$C - AB \ge 0$

If $\theta_1 > 0$, we have

$$\left(\frac{C}{AB} - 1\right)\left(e^{\theta_1(a-a^*)} - 1\right) \geq 0$$

$$\Updownarrow$$

$$e^{\theta_1(a-a^*)} \geq NDE_{OR}$$

From the property that

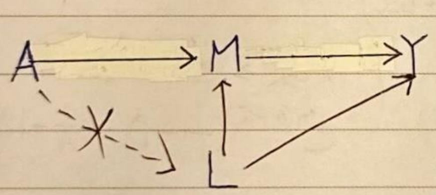
$$TE_{OR} = NIE_{OR} \times NDE_{OR}$$

$$\frac{e^{\beta_1(a-a^*)}}{NDE_{OR}} = NIE_{OR} \geq e^{(\beta_1 - \theta_1)(a-a^*)}$$

(Remark. $CDE_{OR} = e^{\theta_1(a-a^*)} \geq NDE_{OR}$

$PE_{OR} = e^{(\beta_1-\theta_1)(a-a^*)} \leq NIE_{OR}$)

§ Interventional Approach



$\Rightarrow Y(a,m) \perp\!\!\!\perp M(a^*) \mid C$

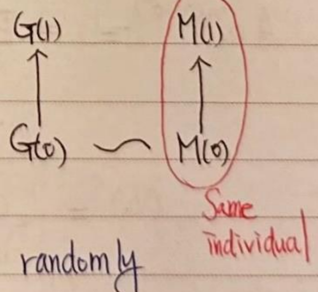
- The cross would exchangeability is fail when exposure affect M-Y confounder.

★ Let G denote random draws from dist^n M

, $G(a)$ is random draws of $M(a)$

1) $G \perp\!\!\!\perp A$

2) $F_G(u) = F_M(u)$, $G \sim M$



Alternative definitions for NDE and NIE is randomly interventional analogues of NDE and NIE

$$IIE = \mathbb{E}[Y(a, G(a))] - \mathbb{E}[Y(a, G(a^*))]$$

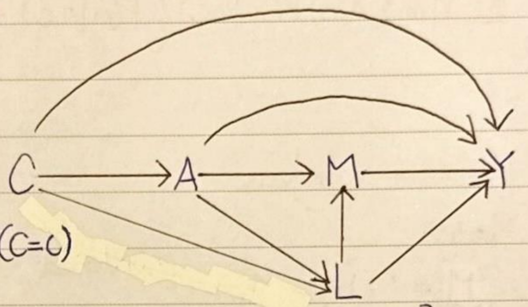
$$IDE = \mathbb{E}[Y(a, G(a^*))] - \mathbb{E}[Y(a^*, G(a^*))]$$

$\psi(a,a^*) = \mathbb{E}[Y(a, G(a^*))]$

$\quad \text{IIE} = \psi(a,a) - \psi(a,a^*)$

$\quad \text{IDE} = \psi(a,a^*) - \psi(a^*,a^*)$

☆ Idenfication of $\psi(a,a^*)$



$$\psi(a,a^*) = \mathbb{E}[Y(a,G(a^*))]$$

$$= \sum_c \mathbb{E}[Y(a,G(a^*)) \mid C=c] P(C=c)$$

$$= \sum_c \left\{ \sum_m \mathbb{E}[Y(a,G(a^*)) \mid G(a^*)=m, C=c] P(G(a^*)=m \mid C=c) \right\} P(C=c)$$

(Consistency) $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid G(a^*)=m, C=c] P(G(a^*)=m \mid C=c) P(C=c)$

$(G(a^*) \perp\!\!\!\perp Y(a,m) \mid C)$ $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid C=c] P(G(a^*)=m \mid C=c) P(C=c)$

$(G(a^*) \sim M(a^*) \mid C)$ $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid C=c] P(M(a^*)=m \mid C=c) P(C=c)$

$(M(a^*) \perp\!\!\!\perp A \mid C)$ $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid C=c] P(M(a^*)=m \mid A=a^*, C=c) P(C=c)$

(Consistency) $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid C=c] P(M=m \mid A=a^*, C=c) P(C=c)$

$(Y(a,m) \perp\!\!\!\perp A \mid C)$ $= \sum_{c,m} \mathbb{E}[Y(a,m) \mid A=a, C=c] P(M=m \mid A=a^*, C=c) P(C=c)$

$$= \sum_{c,m} \left\{ \sum_l \mathbb{E}[Y(a,m) \mid L=l, A=a, C=c] P(L=l \mid A=a, C=c) \right\}$$
$$\times P(M=m \mid A=a^*, C=c) P(C=c)$$

$(Y(a,m) \perp\!\!\!\perp M \mid A, L, C)$

$$= \sum_{c,m,l} \mathbb{E}[Y(a,m) \mid M=m, L=l, A=a, C=c] P(L=l \mid A=a, C=c)$$
$$\times P(M=m \mid A=a^*, C=c) P(C=c)$$

(Consistency)

$$= \sum_{c,m,l} \mathbb{E}(Y \mid M=m, L=l, A=a, C=c) P(L=l \mid A=a, C=c)$$
$$\times P(M=m \mid A=a^*, C=c) P(C=c)$$

$$IDE = \sum_{c,m,l} \left\{ E(Y \mid m,l,a,c)\,P(l \mid a,c) - E(Y \mid m,l,a^*)\,P(l \mid a^*,c) \right\} \times P(m \mid a^*,c)\,P(c)$$

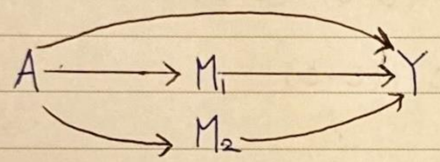
$$IIE = \sum_{c,m,l} E(Y \mid m,l,a,c)\,P(l \mid a,c) \left\{ P(m \mid a,c) - P(m \mid a^*,c) \right\} P(c)$$

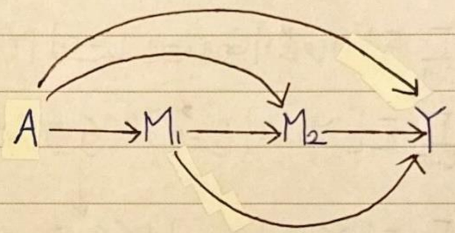
§ Multiple Mediation Analysis

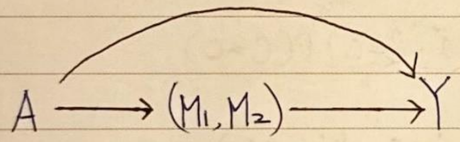
Multiple Mediator

$$\underset{\sim}{M} = (M_1, \ldots, M_p) \begin{cases} \text{Ordered} \\ \text{Non-ordered} \end{cases}$$

DAGs







★ Path Specific Effect (PSE)

Evaluate the effect mediated by a certain combination of Mediators.

In general, k mediators $\Rightarrow 2^k$ PSEs. (increse exponentially)

Ex.	$A \to Y$ (DE)	$(0,0)$	path not mediated by M_1 or M_2
	$A \to M_1 \to Y$	$(1,0)$	path mediated by M_1 only
	$A \to M_2 \to Y$	$(0,1)$	" M_2 "
	$A \to M_1 \to M_2 \to Y$	$(1,1)$	path mediated by M_1 then M_2

$$I = (I(M_1), I(M_2))$$

decompose TE into $\begin{cases} 2^{k-1} \text{ indirect effect} \\ 1 \text{ direct effect} \end{cases}$

$$
\begin{tikzcd}
 & & & \varepsilon_Y \arrow[d] \\
A \arrow[r] \arrow[rr, bend left] \arrow[rrr, bend left=50] & M_1 \arrow[r] \arrow[rr, bend right] & M_2 \arrow[r] & Y \\
\varepsilon_A \arrow[u] & \varepsilon_1 \arrow[u] & \varepsilon_2 \arrow[u] &
\end{tikzcd}
$$

$$A = g_A(\varepsilon_A)$$
$$M_1 = g_1(A, \varepsilon_1)$$
$$M_2 = g_2(A, M_1, \varepsilon_2)$$
$$Y = g_Y(A, M_1, M_2, \varepsilon_Y)$$
$$\Downarrow$$
$$A = 1$$
$$M_1(1) = g_1(1, \varepsilon_1)$$
$$M_2(1) = M_2(1, M_1(1)) = g_2(1, M_1(1), \varepsilon_2)$$
$$Y(1) = Y(1, M_1(1), M_2(1, M_1(1))) = g_Y(1, M_1(1), M_2(1, M_1(1)), \varepsilon_Y)$$

$$
\begin{aligned}
TE &= Y(1) - Y(0) \\
&= Y(1, M_1(1), M_2(1, M_1(1))) \\
&\quad - Y(0, M_1(0), M_2(0, M_1(0)))
\end{aligned}
\qquad
\left(
\begin{array}{lll}
\longrightarrow & \text{for path} & A \to Y \\
\dashrightarrow \text{(dotted)} & \text{"} & A \to M_1 \to Y \\
\dashrightarrow & \text{"} & A \to M_2 \to Y \\
\dashrightarrow \text{(dash-dot)} & \text{"} & A \to M_1 \to M_2 \to Y
\end{array}
\right)
$$

$$
\begin{aligned}
&= \left[Y(1, M_1(1), M_2(1, M_1(1))) - Y(1, M_1(1), M_2(1, M_1(0))) \right] \iff PSE_{12} \\
&+ \left[Y(1, M_1(1), M_2(1, M_1(0))) - Y(1, M_1(1), M_2(0, M_1(0))) \right] \iff PSE_2 \\
&+ \left[Y(1, M_1(1), M_2(0, M_1(0))) - Y(1, M_1(0), M_2(0, M_1(0))) \right] \iff PSE_1 \\
&+ \left[Y(1, M_1(0), M_2(0, M_1(0))) - Y(0, M_1(0), M_2(0, M_1(0))) \right] \iff PSE_0
\end{aligned}
$$

Decomposition Strategies

1. Two-Way Decomp.
2. Parallel Decomp.
3. Sequential Decomp
4. Partial Backard/Forward Decomp
5. Complete Decomp.

☆ Two-Way decomposition

All mediators are treated as one multivariate mediator

$$\boldsymbol{M} = (M_1, \dots, M_p)$$

TE is decomposed into $\begin{cases} \text{passing through } \boldsymbol{M} \\ \text{not passing through } \boldsymbol{M} \end{cases}$

$\Rightarrow$

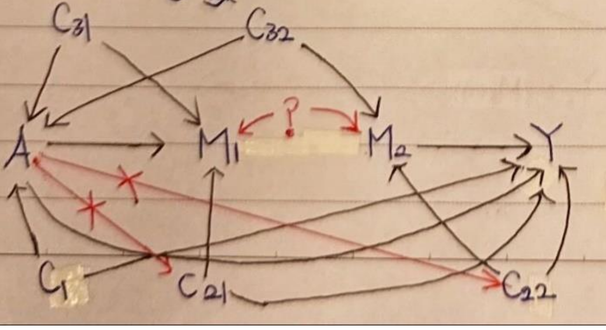
$$\begin{aligned} TE &= \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] \\ &= \mathbb{E}[Y(1, \boldsymbol{M}(1))] - \mathbb{E}[Y(0, \boldsymbol{M}(0))] \\ &= \Big[\mathbb{E}[Y(1, \boldsymbol{M}(1))] - \mathbb{E}[Y(1, \boldsymbol{M}(0))]\Big] \Leftrightarrow NIE \\ &\quad + \Big[\mathbb{E}[Y(1, \boldsymbol{M}(0))] - \mathbb{E}[Y(0, \boldsymbol{M}(0))]\Big] \Leftrightarrow NDE \end{aligned}$$

Define $\phi_{TW}(a, a^*) = \mathbb{E}[Y(a, \boldsymbol{M}(a^*))]$

$$\begin{aligned} TE &= \phi_{TW}(1,1) - \phi_{TW}(1,0) + \phi_{TW}(1,0) - \phi_{TW}(0,0) \\ &= NIE_{TW} + NDE_{TW} \end{aligned}$$

Identification of $\phi_{TW}(a, a^*)$ is

$$\sum_c \sum_{\boldsymbol{m}} \mathbb{E}(Y \mid \boldsymbol{M} = \boldsymbol{m}, A = a, C = c)\, P(\boldsymbol{M} = \boldsymbol{m} \mid A = a^*, C = c)\, P(C = c)$$



$Y(a, \boldsymbol{m}) \perp\!\!\!\perp A \mid C \to C_1$

$Y(a, \boldsymbol{m}) \perp\!\!\!\perp \boldsymbol{M} \mid A, C \to (C_{21}, C_{22})$

$\boldsymbol{M}(a) \perp\!\!\!\perp A \mid C \to (C_{31}, C_{32})$

$Y(a, \boldsymbol{m}) \perp\!\!\!\perp \boldsymbol{M}(a^*) \mid C$